```
#Assignment 1: nstall and configure TensorFlow/Keras in Google Colab. Perform data preprocessing,
#normalization, train-test splitting, and data visualization on a sample dataset.
```

```
import tensorflow as tf
import keras

print("TensorFlow Version:", tf.__version__)
print("Keras Version:", keras.__version__)
```

```
TensorFlow Version: 2.20.0
Keras Version: 3.13.2
```

```
#import libraries

import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

print("Libraries Imported Successfully!")
```

```
Libraries Imported Successfully!
```

```
#load dataset

diabetes = load_diabetes()

X = diabetes.data
y = diabetes.target

df = pd.DataFrame(X, columns=diabetes.feature_names)

df["Target"] = y

print("Dataset Loaded Successfully!")

df.head()
```

```
Dataset Loaded Successfully!
```

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	Target
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019907	-0.017646	151.0
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068332	-0.092204	75.0
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592	0.002861	-0.025930	141.0
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022688	-0.009362	206.0
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031988	-0.046641	135.0

```
print("Shape of Dataset:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

```
Shape of Dataset:
(442, 11)

Column Names:
```

```
Index(['age', 'sex', 'bmi', 'bp', 's1', 's2', 's3', 's4', 's5', 's6',
       'Target'],
      dtype='object')

Missing Values:
age       0
sex       0
bmi       0
bp        0
s1        0
s2        0
s3        0
s4        0
s5        0
s6        0
Target    0
dtype: int64

Duplicate Rows:
0
```

```
df.describe()
```

	age	sex	bmi	bp	s1	s2	s3	s4	s5
count	4.420000e+02	4.420000e+02	4.420000e+02	4.420000e+02	4.420000e+02	4.420000e+02	4.420000e+02	4.420000e+02	4.420000e+02
mean	-2.511817e-19	1.230790e-17	-2.245564e-16	-4.797570e-17	-1.381499e-17	3.918434e-17	-5.777179e-18	-9.042540e-18	9.293722e-17
std	4.761905e-02	4.761905e-02	4.761905e-02	4.761905e-02	4.761905e-02	4.761905e-02	4.761905e-02	4.761905e-02	4.761905e-02
min	-1.072256e-01	-4.464164e-02	-9.027530e-02	-1.123988e-01	-1.267807e-01	-1.156131e-01	-1.023071e-01	-7.639450e-02	-1.260971e-01
25%	-3.729927e-02	-4.464164e-02	-3.422907e-02	-3.665608e-02	-3.424784e-02	-3.035840e-02	-3.511716e-02	-3.949338e-02	-3.324559e-02
50%	5.383060e-03	-4.464164e-02	-7.283766e-03	-5.670422e-03	-4.320866e-03	-3.819065e-03	-6.584468e-03	-2.592262e-03	-1.947171e-03

```
#data normalization

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

print("Data Normalized Successfully!")

print(X_scaled[:5])
```

```
Data Normalized Successfully!
[[ 0.80050009  1.06548848  1.29708846  0.45984057 -0.92974581 -0.73206462
  -0.91245053 -0.05449919  0.41853093 -0.37098854]
 [-0.03956713 -0.93853666 -1.08218016 -0.55350458 -0.17762425 -0.40288615
   1.56441355 -0.83030083 -1.43658851 -1.93847913]
 [ 1.79330681  1.06548848  0.93453324 -0.1192138  -0.95867356 -0.71889748
  -0.68024452 -0.05449919  0.06015558 -0.54515416]
 [-1.87244107 -0.93853666 -0.24377122 -0.77064997  0.25629203  0.52539714
  -0.75764652  0.72130245  0.47698252 -0.19682291]
 [ 0.11317236 -0.93853666 -0.76494435  0.45984057  0.08272552  0.32789006
   0.17117751 -0.05449919 -0.67250161 -0.98056821]]
```

```
#train test spliting

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y,
    test_size=0.20,
    random_state=42
)

print("Training Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
```

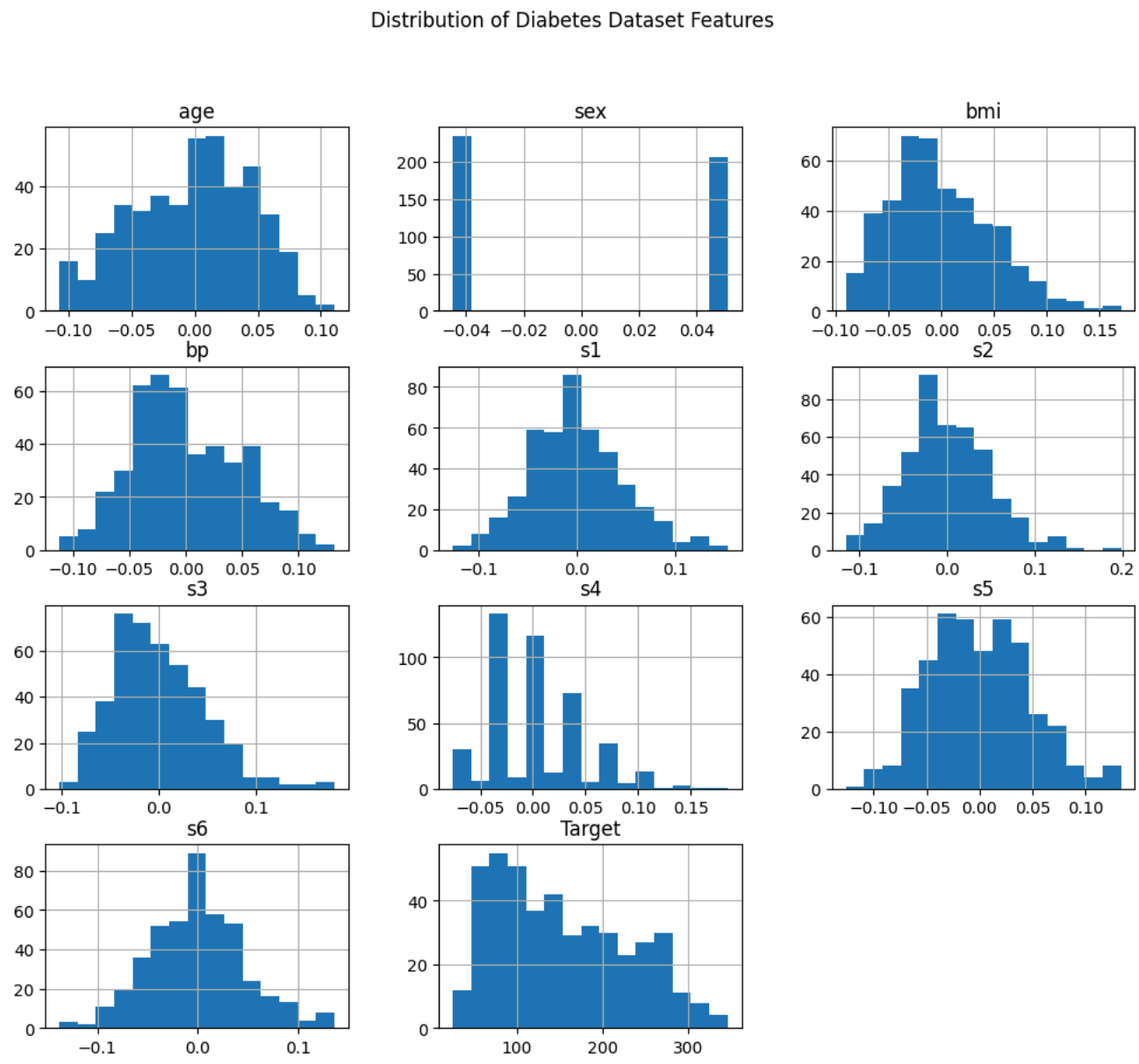
```
Training Data Shape: (353, 10)
Testing Data Shape: (89, 10)
```

```
#histogram

df.hist(figsize=(12,10), bins=15)

plt.suptitle("Distribution of Diabetes Dataset Features")

plt.show()
```



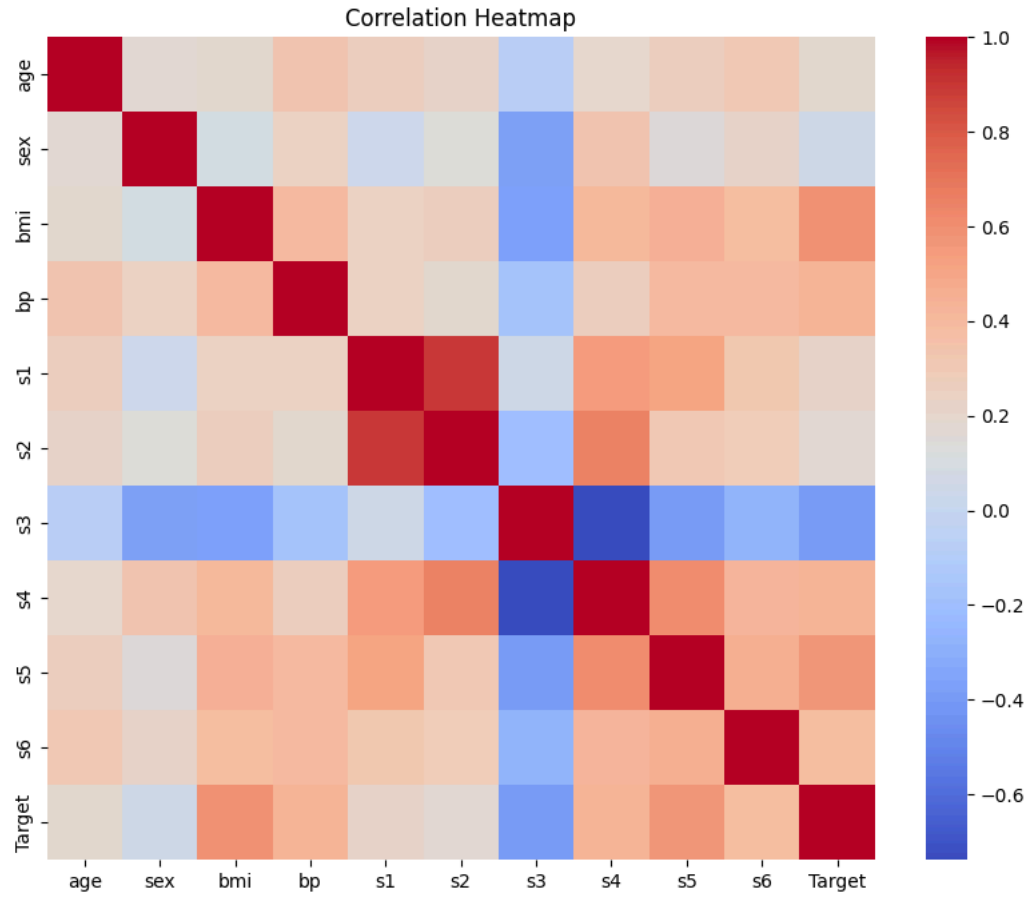

```
#correlation heatmap

plt.figure(figsize=(10,8))

sns.heatmap(
    df.corr(),
    cmap="coolwarm",
    annot=False
)

plt.title("Correlation Heatmap")

plt.show()
```




```
#box plot

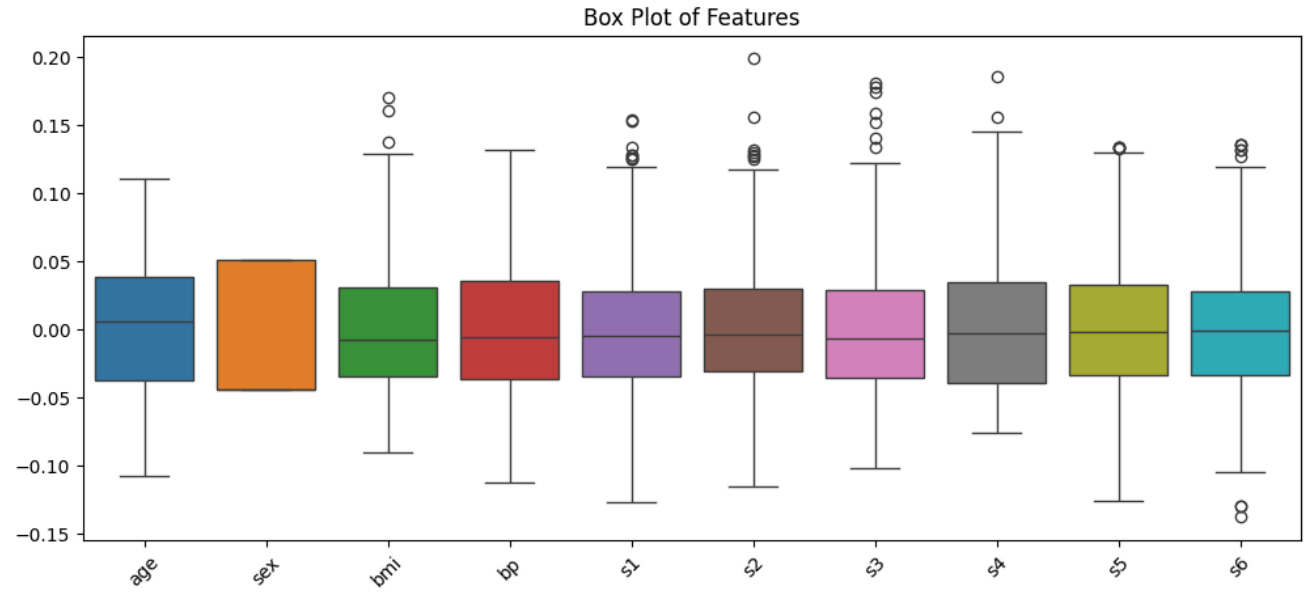
plt.figure(figsize=(12,5))

sns.boxplot(data=df.iloc[:, :-1])

plt.title("Box Plot of Features")

plt.xticks(rotation=45)

plt.show()
```



```
#scatter plot

plt.figure(figsize=(7,5))

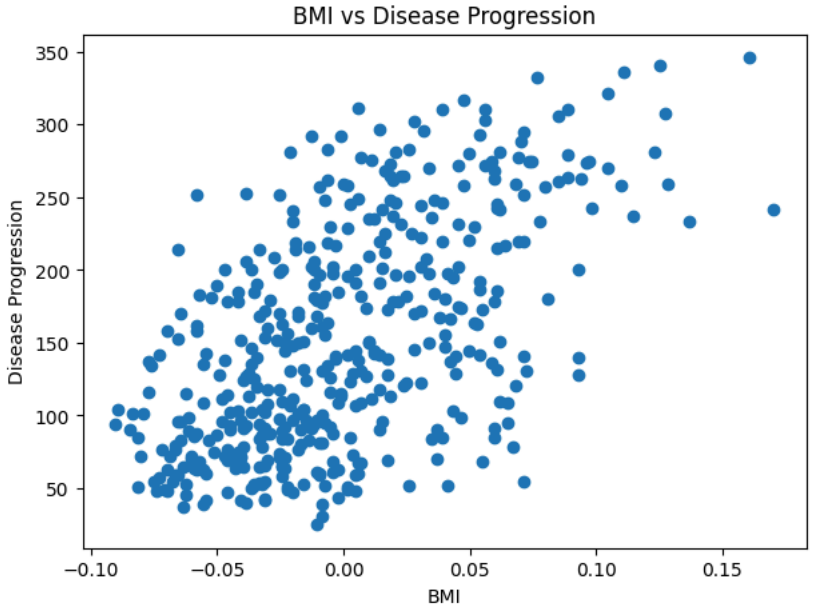
plt.scatter(df["bmi"], df["Target"])

plt.xlabel("BMI")

plt.ylabel("Disease Progression")

plt.title("BMI vs Disease Progression")

plt.show()
```

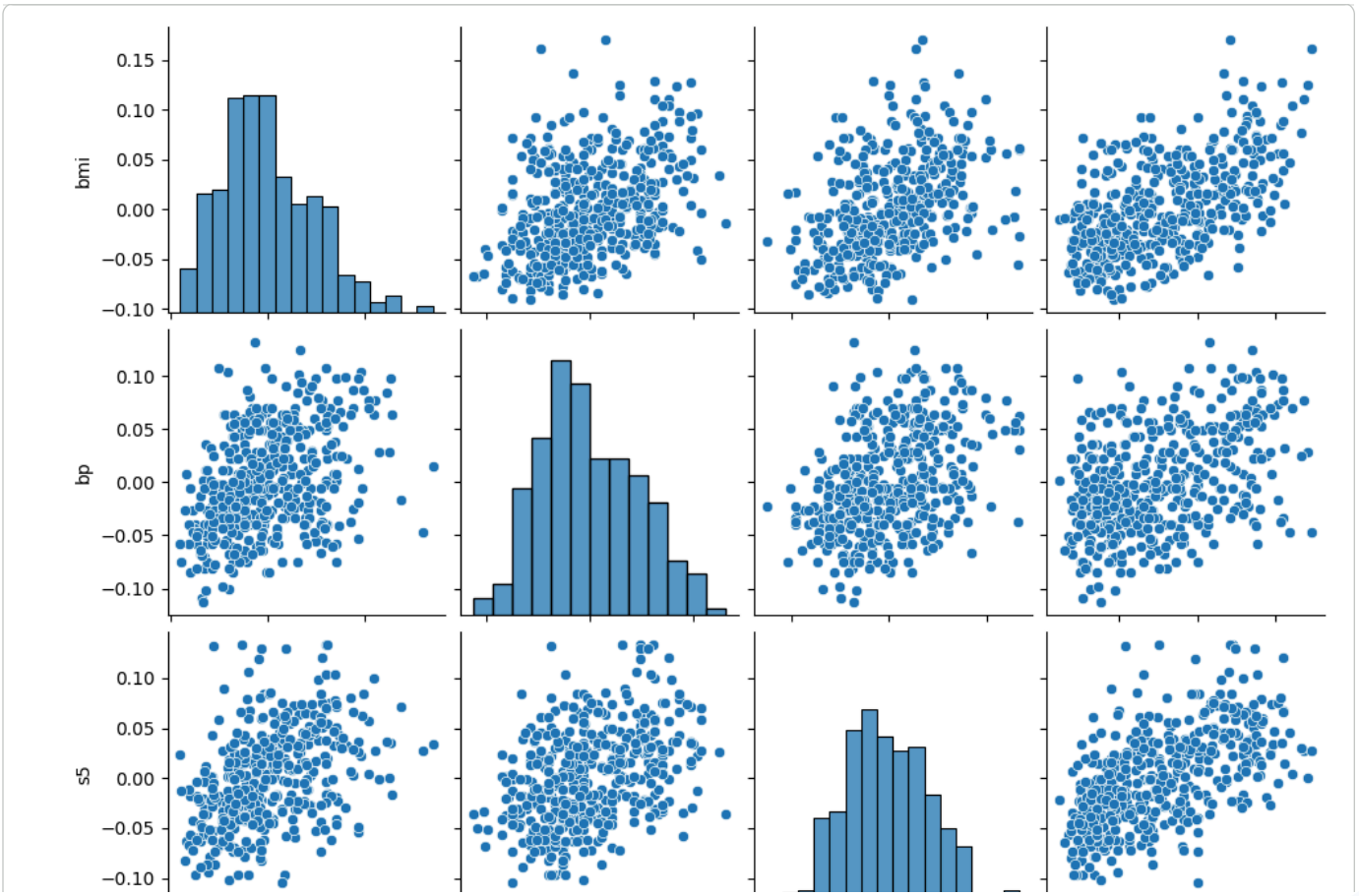



```
#pair plot

small_df = df[["bmi", "bp", "s5", "Target"]]

sns.pairplot(small_df)

plt.show()
```


bmi
bp
s5
0.15
0.10
0.05
0.00
−0.05
−0.10